Fast Probabilistic Algorithms for Proving Pairings

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October 2, 2025

In this paper, we introduce a public-coin interactive proof system for efficiently proving the correctness of elliptic curve pairing computations. Elliptic curve pairings form the mathematical foundation of many modern cryptographic protocols, yet their verification remains computationally intensive.

In resource-constrained environments—including BitcoinVM, blockchain smart contracts, ZKVMs, and recursive zero-knowledge proof systems—traditional pairing verification remains expensive. This limits the applicability and affordability of privacy-preserving applications, effectively hindering widespread adoption of blockchain technology.

We systematically present a progression of state-of-the-art techniques: El Housni's foundational R1CS pairing optimizations [Hou22], Feltroidprime's enhanced extension field arithmetic [Fel23], and Novakovic and Eagen's final exponentiation elimination [NE24].

Building upon these foundations, our approach consolidates Miller loop iterations and final exponentiation verification into univariate polynomial relations that can be verified through efficient polynomial identity testing. By verifying these polynomial relationships instead of performing the computations we achieve a 50% improvement in Pairing performance. While our method requires additional auxiliary data compared to traditional pairing verification, it achieves a 75% reduction in commitment overhead relative to Feltroidprime's extension field techniques.

1 Introduction

Blockchain technology is ready for widespread adoption—we have reached a scale where processing Visa-like volumes on-chain is within reach. However, privacy remains a significant barrier to entry for many users and applications. Zero-knowledge proofs (ZKPs) have emerged as a powerful solution to this challenge.

1.1 Motivation

Elliptic curve pairings represent fundamental building blocks for numerous cryptographic systems including Groth16 [Gro16], PLONK [GWC19], BLS signature schemes [BGW+22], and KZG polynomial commitment schemes [KZG10].

However, **pairings are computationally intensive**. In execution environments such as blockchain virtual machines, recursive proof systems, and arithmetized circuits, the costs associated with this computation become prohibitive for practical deployment.

1.2 Our Contribution

We present a novel optimization for pairing computation through a public-coin interactive proof system that consolidates the pairing operations into unified polynomial verifications under the Schwartz-Zippel lemma. By treating complete loop iterations as single polynomial relationships rather than sequences of individual field operations [Fel23], our approach achieves:

- **50% reduction** in the pairing computation complexity
- **Over 70% decrease** in auxiliary data overhead
- Practical deployment enabling ZK proof verification within blockchain transaction limits

Our key insight is that instead of optimizing individual field operations, we can treat complete Miller loop iterations as single polynomial relationships, dramatically reducing both computational and communication costs.

Implementation and Impact: Garaga [FE], a state-of-the-art elliptic curve tooling library for Cairo and Starknet, implements our two round interactive proof system with polynomial batching. These techniques allowed Garaga to fit their pairing transaction within Starknet's 4000 calldata limit while minimising the gas costs.

1.3 Paper organization

The remainder of this paper is organized as follows.

Section 2 provides necessary background on pairings and constraint systems.

Section 3 surveys the evolution of optimization techniques that form our foundation.

Section 4 presents our unified Miller loop optimization framework.

Section 5 outlines the public-coin interactive proof systems with one, two and three rounds for different applications.

Section 6 concludes with implications and future work.

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